Assignment 6: CI/CD Pipeline Implementation

Objective: Build an automated CI/CD pipeline using GitHub Actions for continuous integration and deployment.

GitHub Repository

Repository Name: assignment6-cicd

Repository Structure

assignment6-cicd/
■■■ .github/
■ ■■■ workflows/
■ ■■■ ci-cd.yml
■■■ app/
■ ■■■ Dockerfile
■ ■■■ index.html
■ ■■■ nginx.conf
■■■ screenshots/
■■■ docs/
■ ■■■ deployment-workflow.md
■ ■■■ troubleshooting.md
■ ■■■ rollback-procedure.md
■■■ README.md

Implementation Steps

1. Created GitHub repository for CI/CD implementation.
2. Configured GitHub Actions workflow.
3. Added automated build stage.
4. Added Docker image build process.
5. Configured image push to container registry.
6. Added automatic deployment stage after successful build.
7. Validated pipeline execution using GitHub Actions.

Pipeline Configuration

name: CI/CD Pipeline on: push: branches: [main] jobs: build: runs-on: ubuntu-latest steps: - uses: actions/checkout@v4 - name: Build Docker Image run: docker build -t myapp . - name: Push Image run: docker push myapp - name: Deploy run: echo Deployment Successful

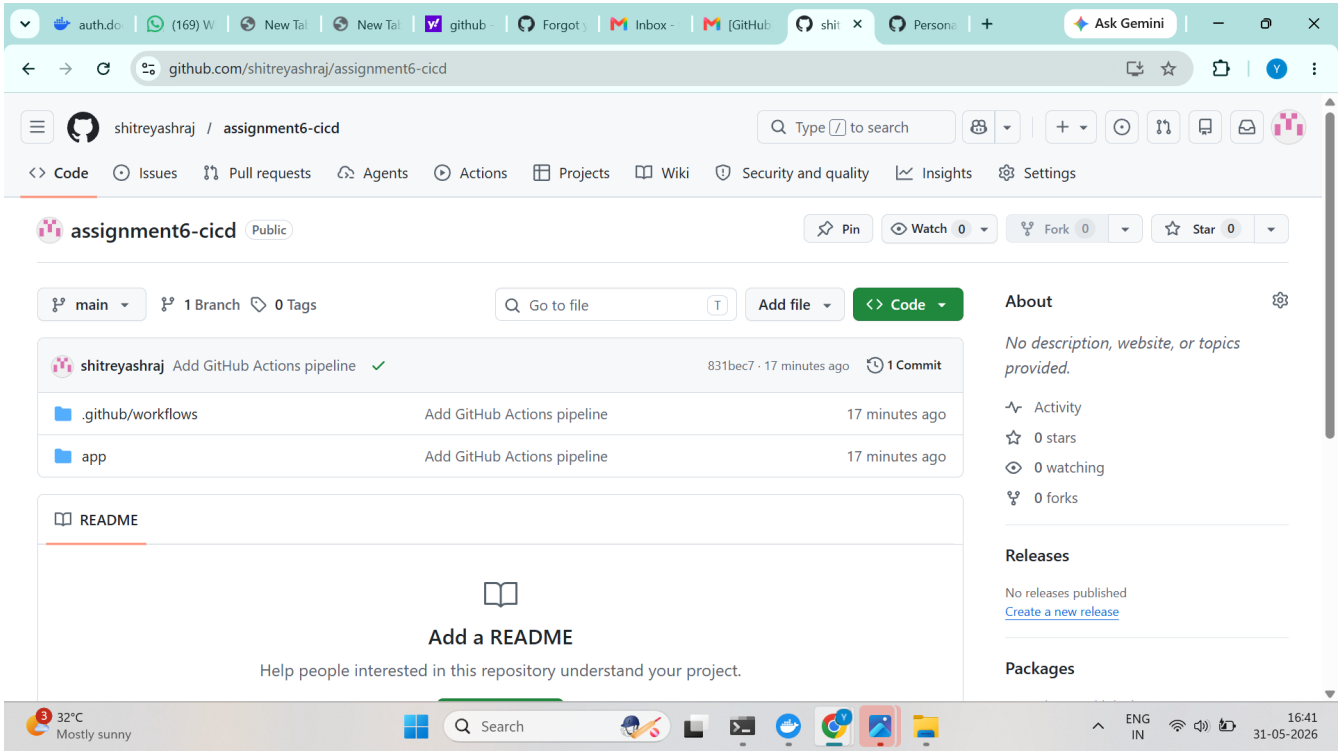
Deployment Workflow

Developer Push Code ↓ GitHub Actions Triggered ↓ Run Tests ↓ Build Docker Image ↓ Push to Registry ↓ Deploy Application ↓ Verify Deployment

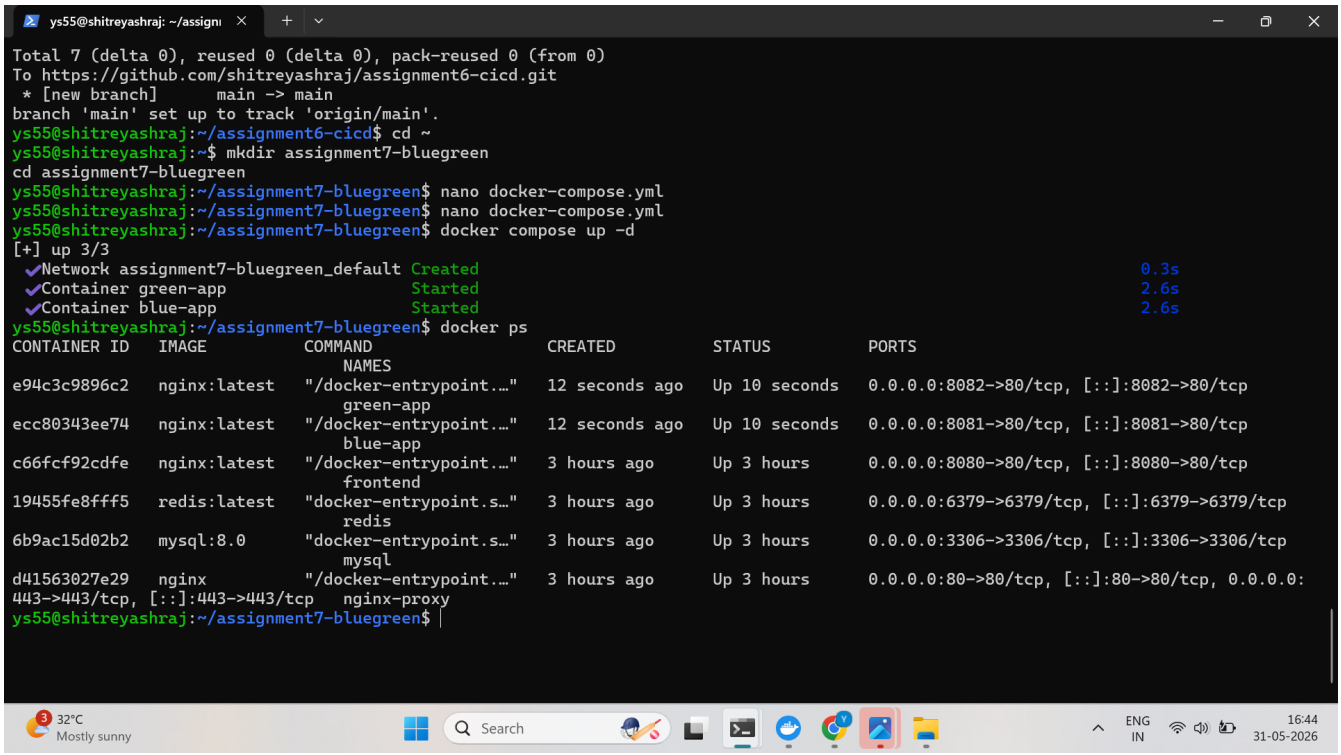
Secrets Management

GitHub Secrets were used to securely store: • Docker Hub Username • Docker Hub Access Token • Deployment Credentials Secrets are referenced inside GitHub Actions and never stored in source code.

Screenshot Evidence



Screenshot Evidence



Troubleshooting Guide

Issue: Workflow not triggered.
Resolution: Verify branch name and workflow syntax.

Issue: Docker build failure.
Resolution: Check Dockerfile and dependencies.

Issue: Authentication failure while pushing image.
Resolution: Verify GitHub Secrets and registry credentials.

Rollback Procedure

1. Revert to previous Git commit. 2. Trigger CI/CD pipeline again. 3. Deploy previously stable Docker image. 4. Verify application health.

Key Decisions and Configuration Explanation

GitHub Actions was selected because it integrates directly with GitHub repositories. Docker images ensure deployment consistency across environments. GitHub Secrets provide secure credential management. Automated deployment reduces manual effort and minimizes human error.